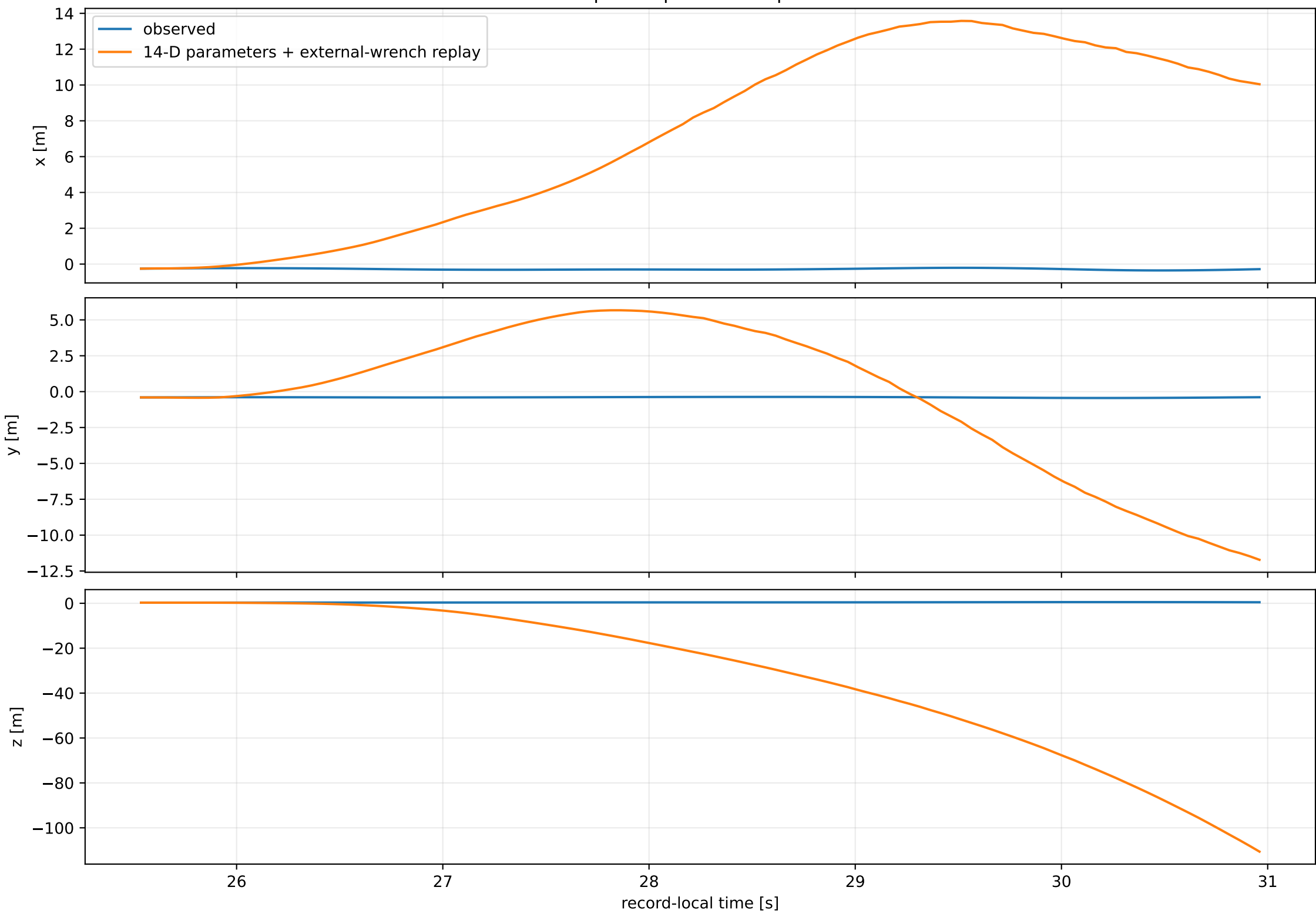
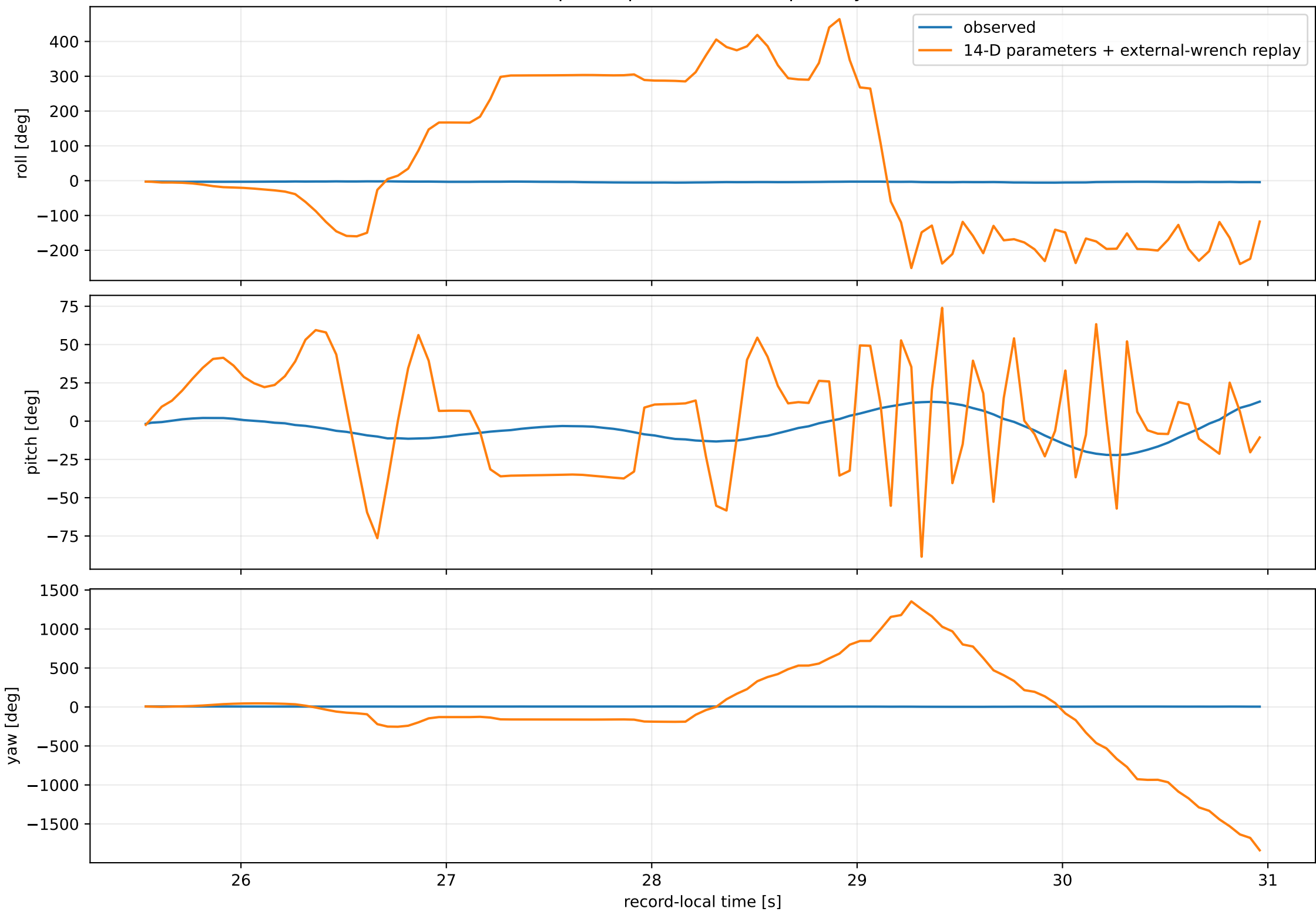


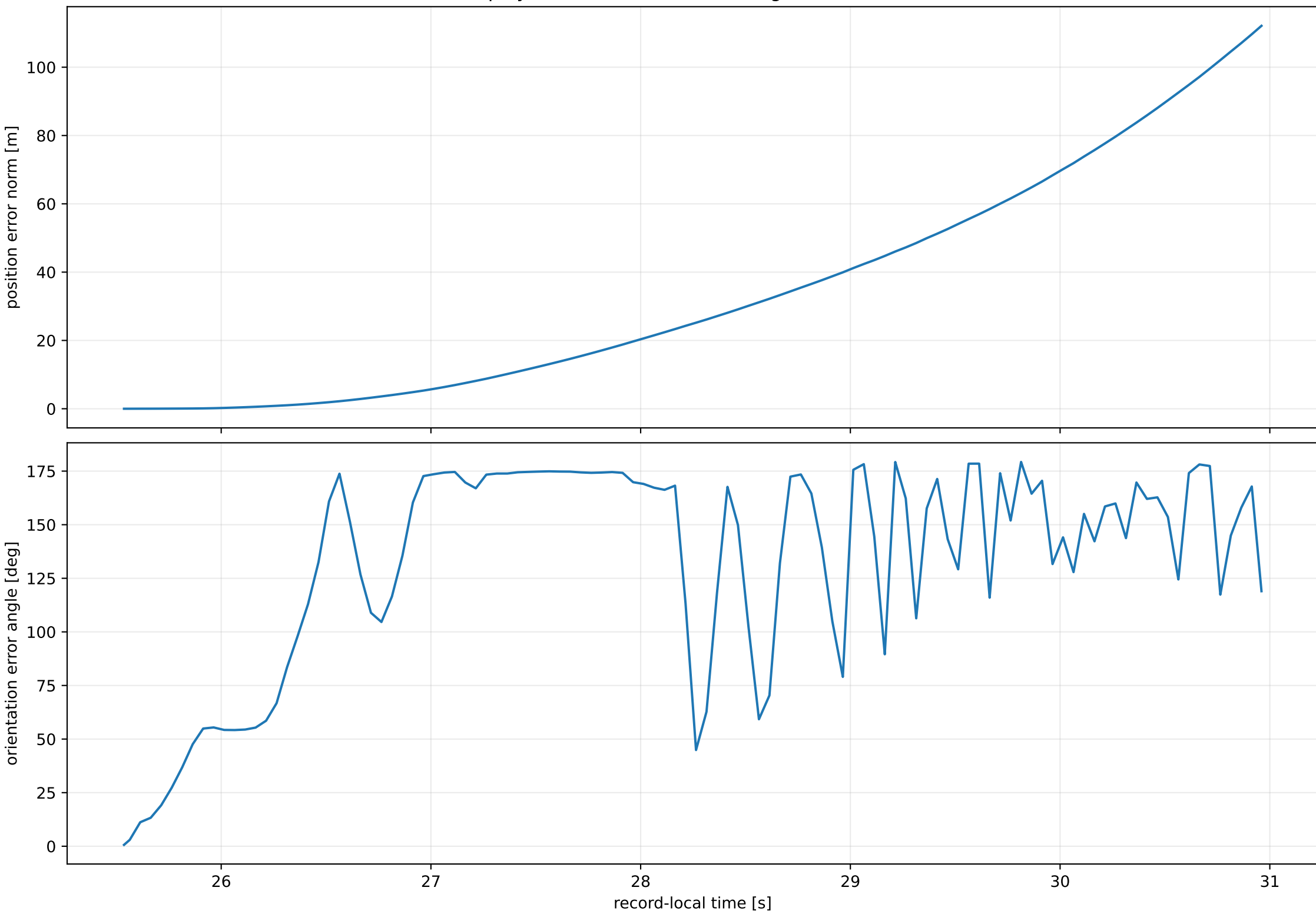
Fine pose reproduction: position



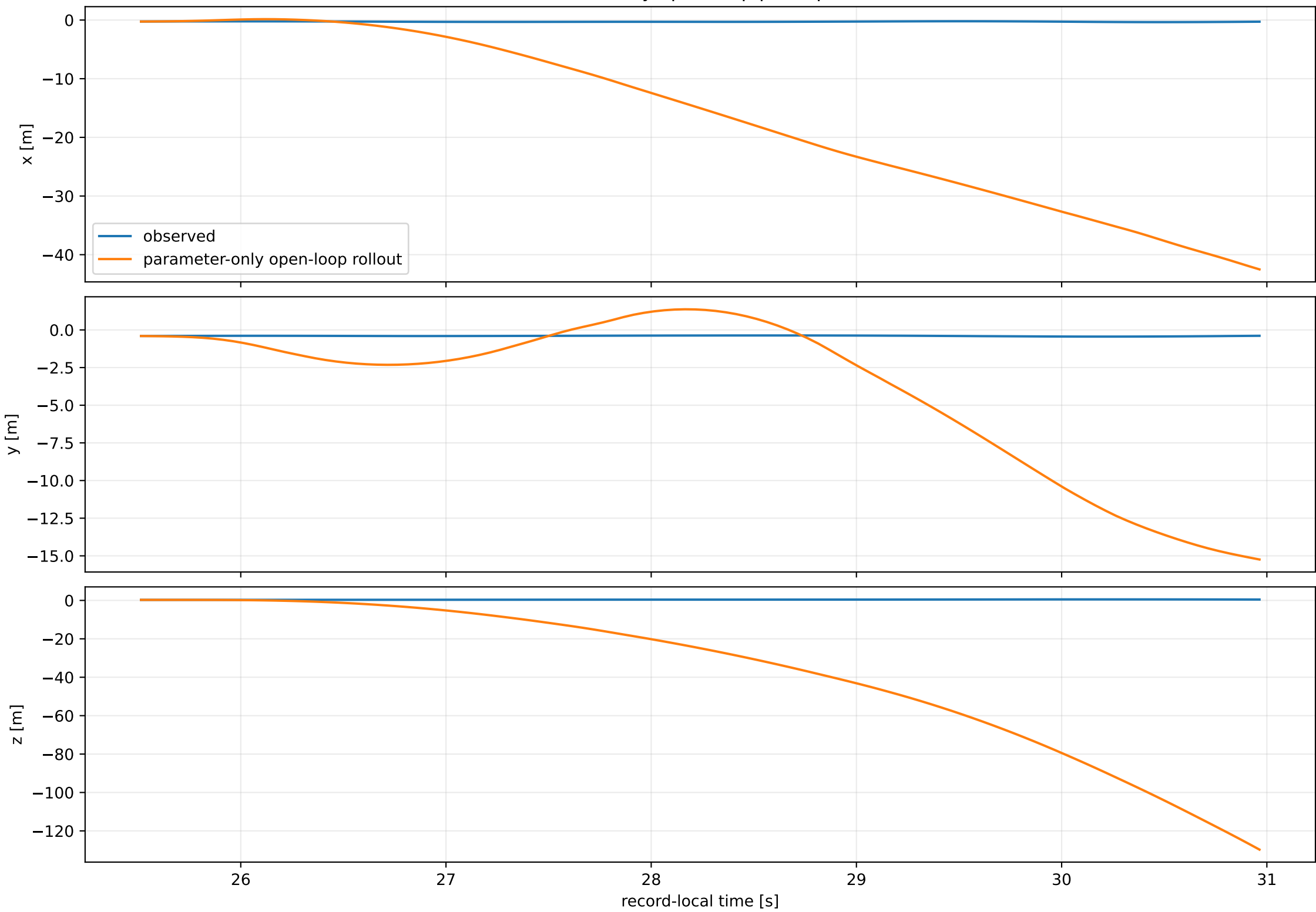
Fine pose reproduction: roll / pitch / yaw



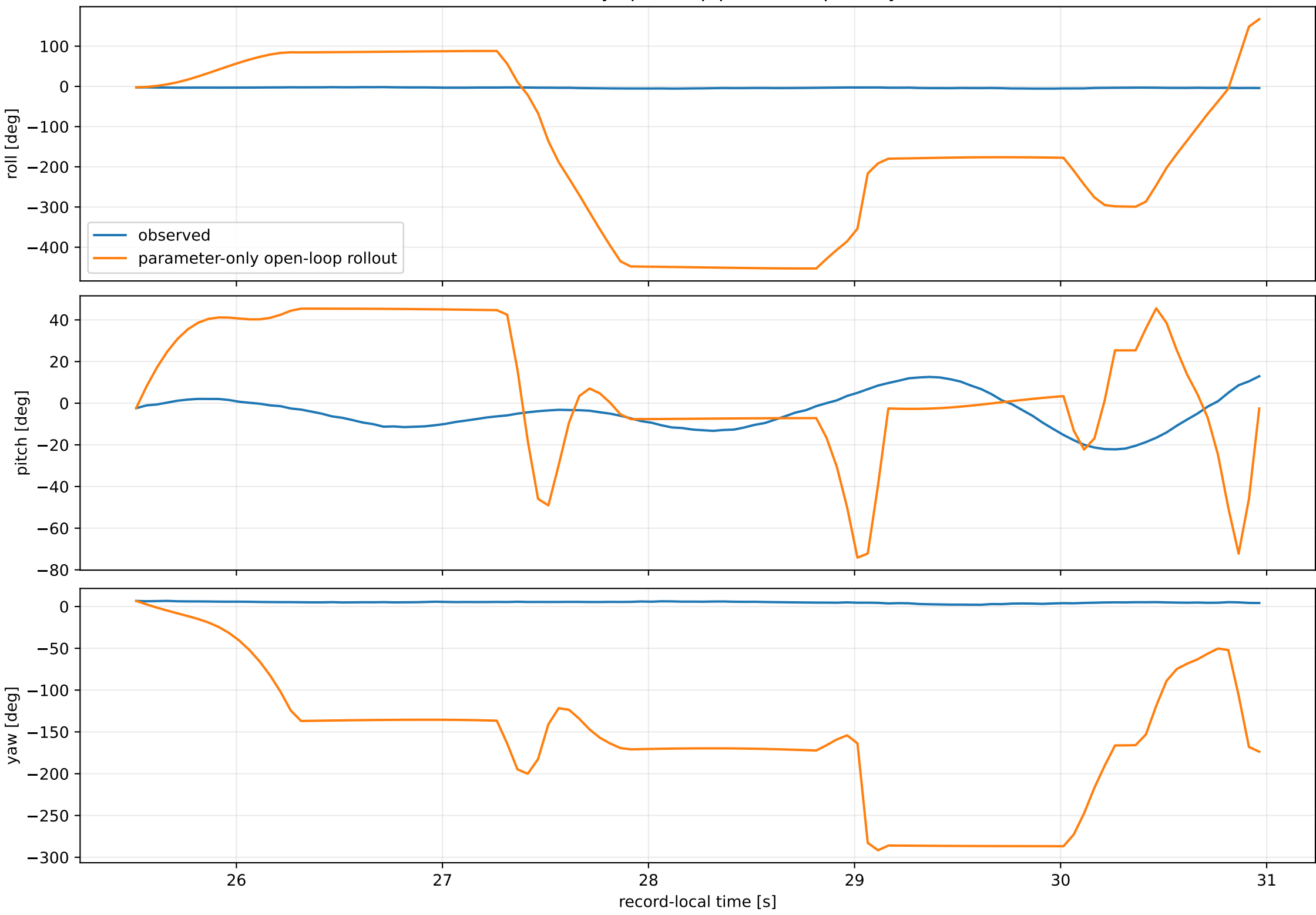
Fine replay error after the 14-D change — lower is better



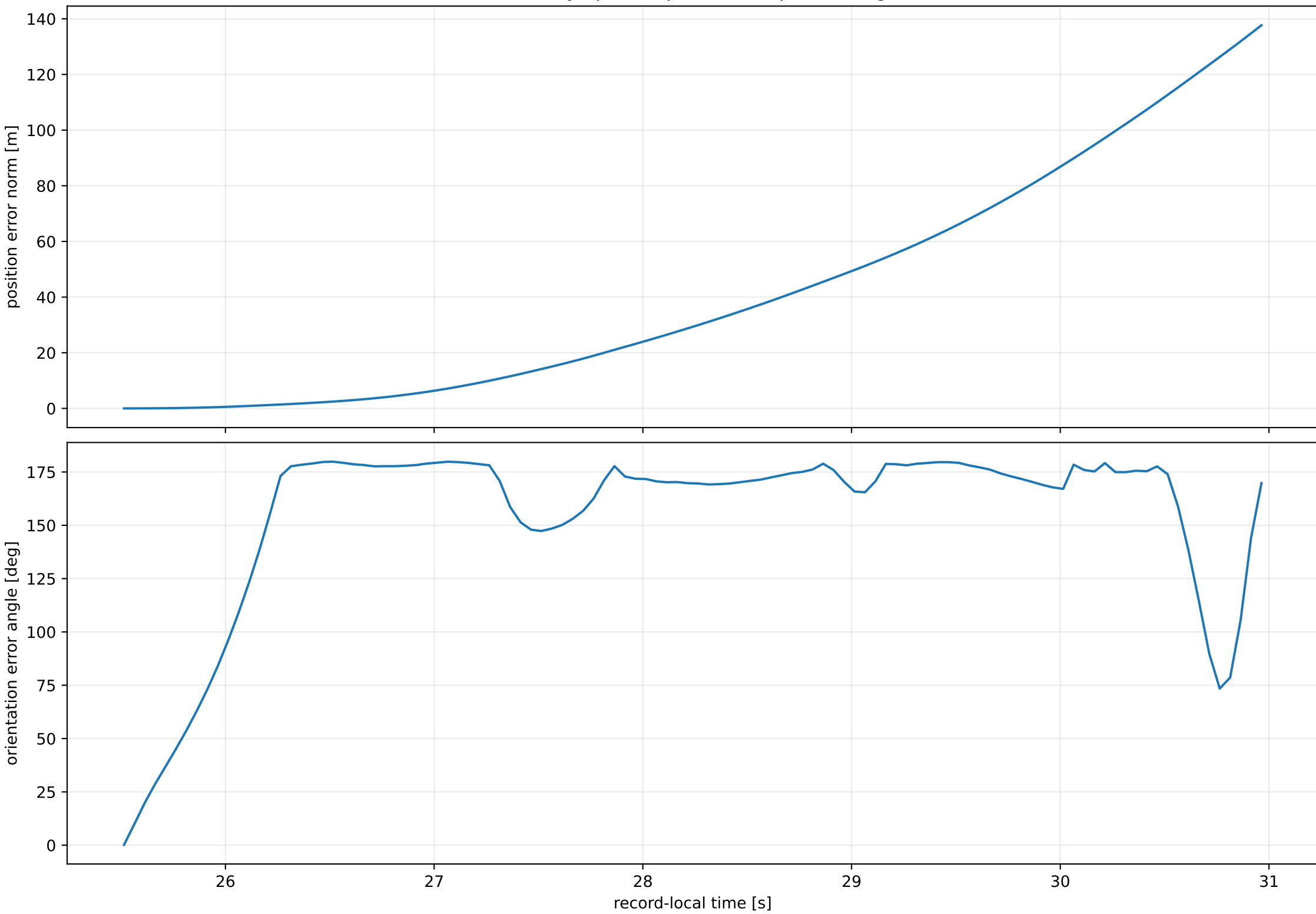
Parameter-only open-loop pose: position



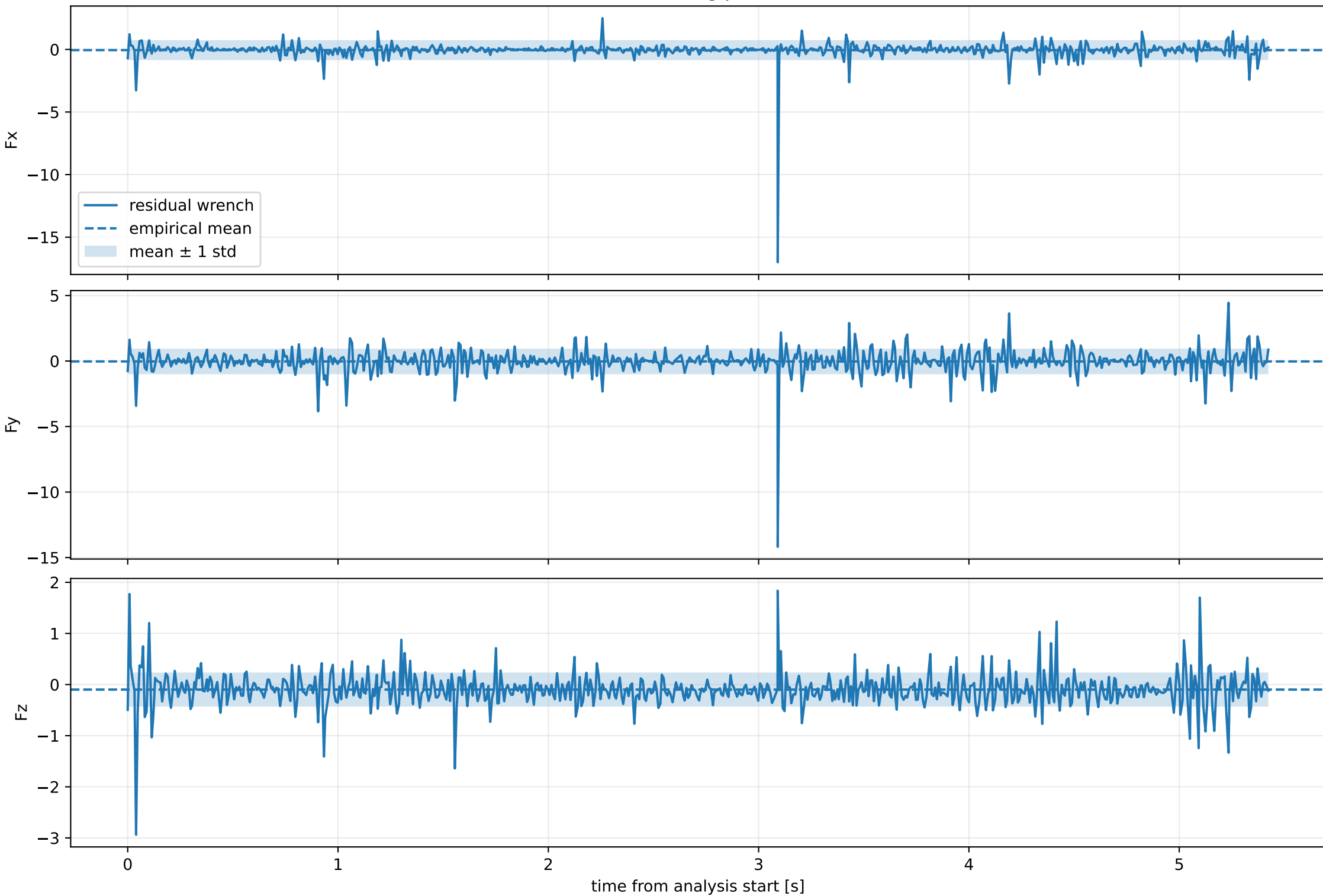
Parameter-only open-loop pose: roll / pitch / yaw



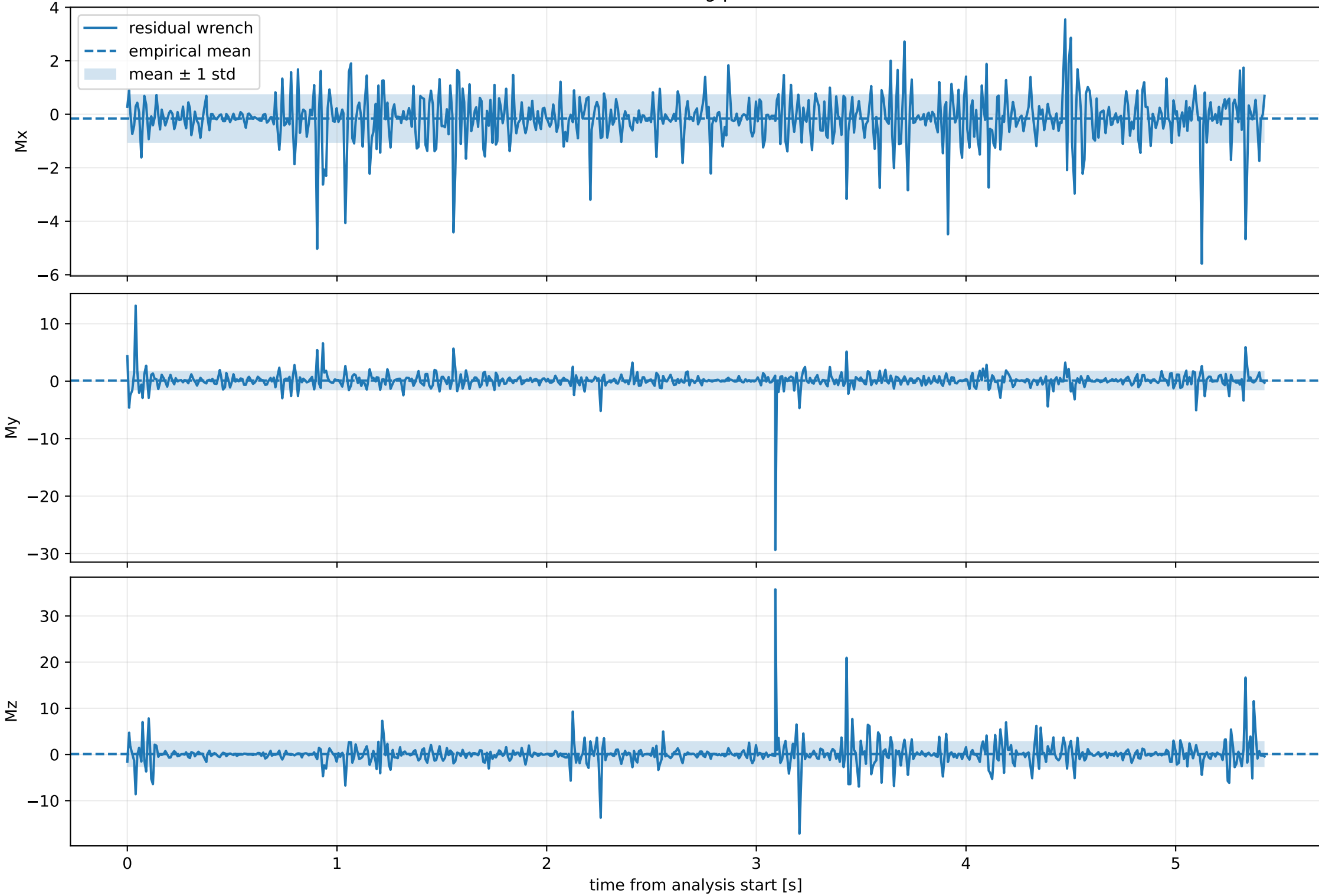
Parameter-only open-loop rollout (separate diagnostic)



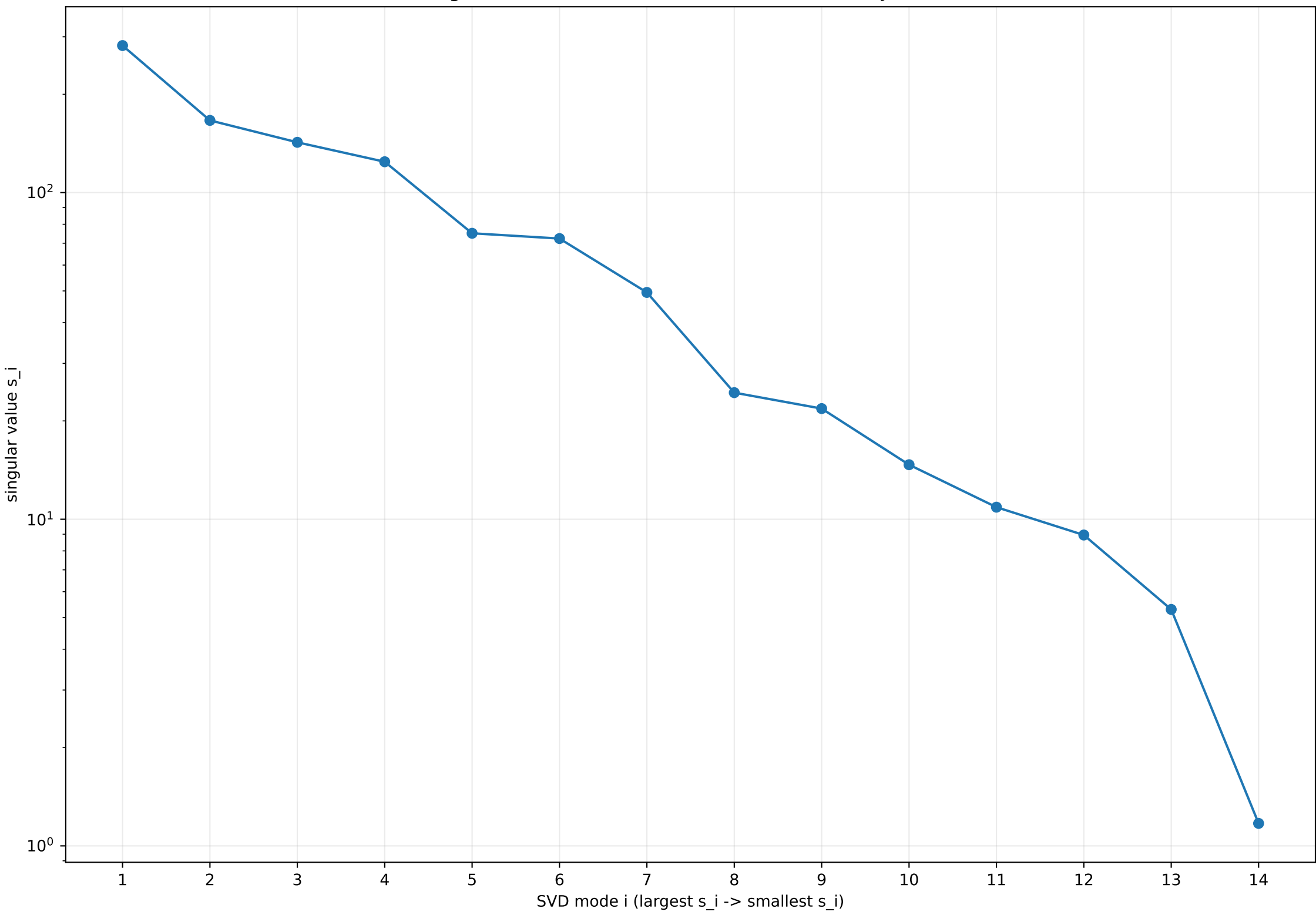
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Decoded physical parameters at the selected estimate

mass [kg]

2.36322838006

inertia J [kg m²]

[0.065995222 -0.00085918275 -0.00085393278]
[-0.00085918275 0.065555349 0.00037661064]
[-0.00085393278 0.00037661064 0.12897435]

principal moments [kg m²]

[0.064887596, 0.066649023, 0.1289883]

CoG position in body frame [m]

[-0.033817774, -0.03721546, 0.011641507]

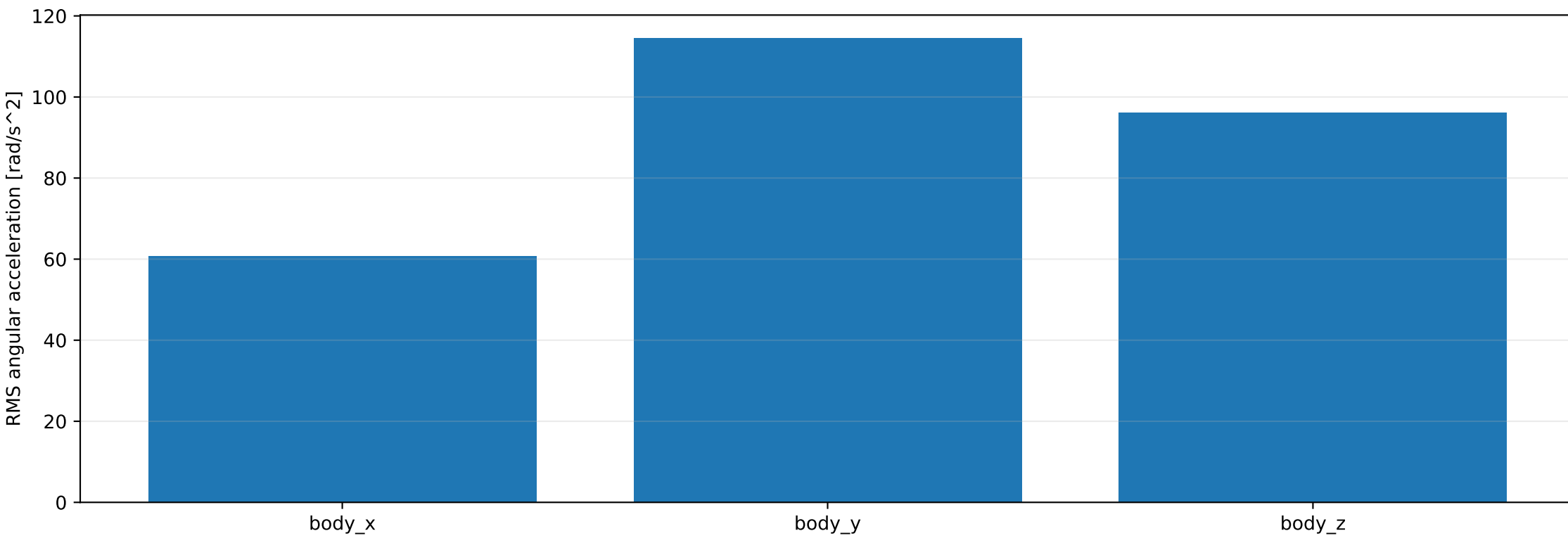
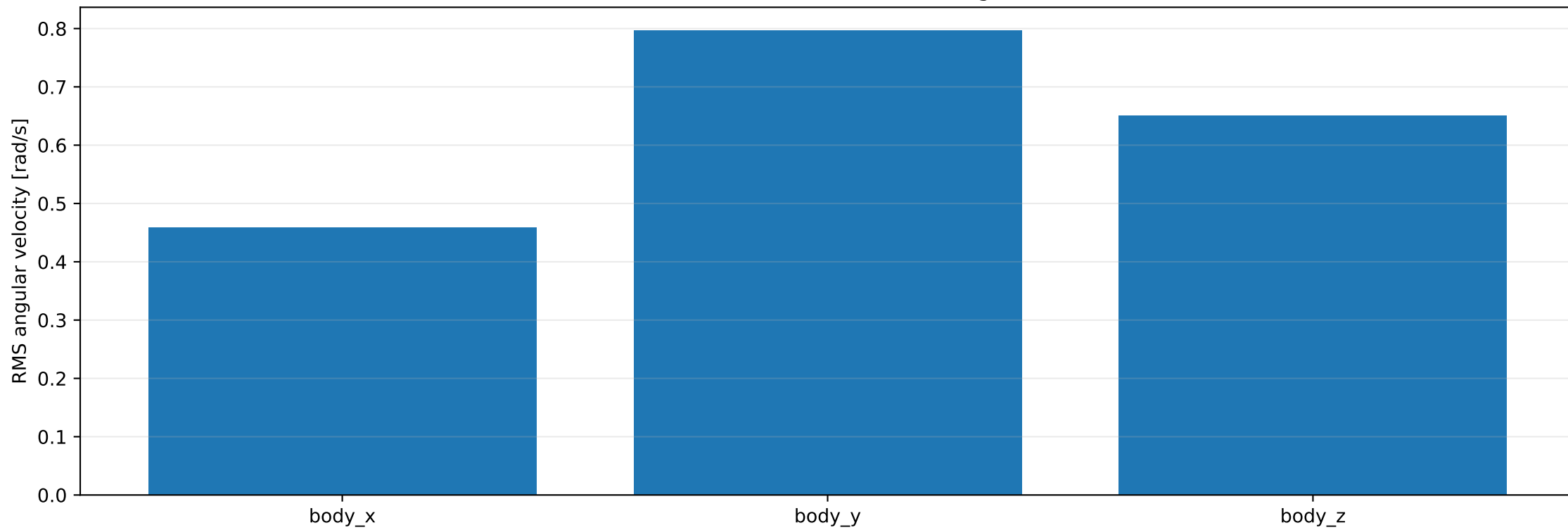
rotor force effectiveness

[1.062632, 0.97105174, 0.85063767, 0.93226868]

selected delay [s]

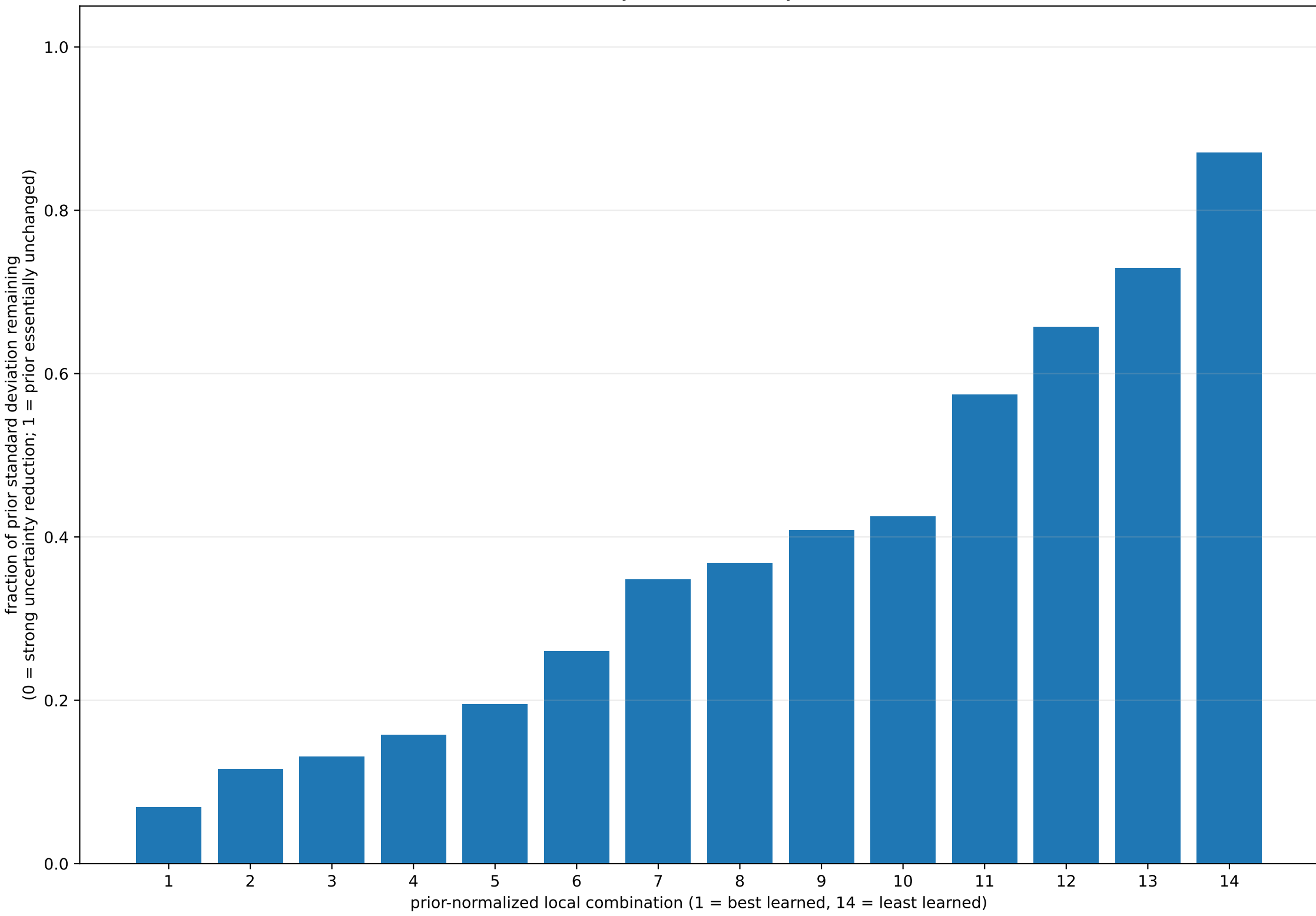
0.051063551966

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

